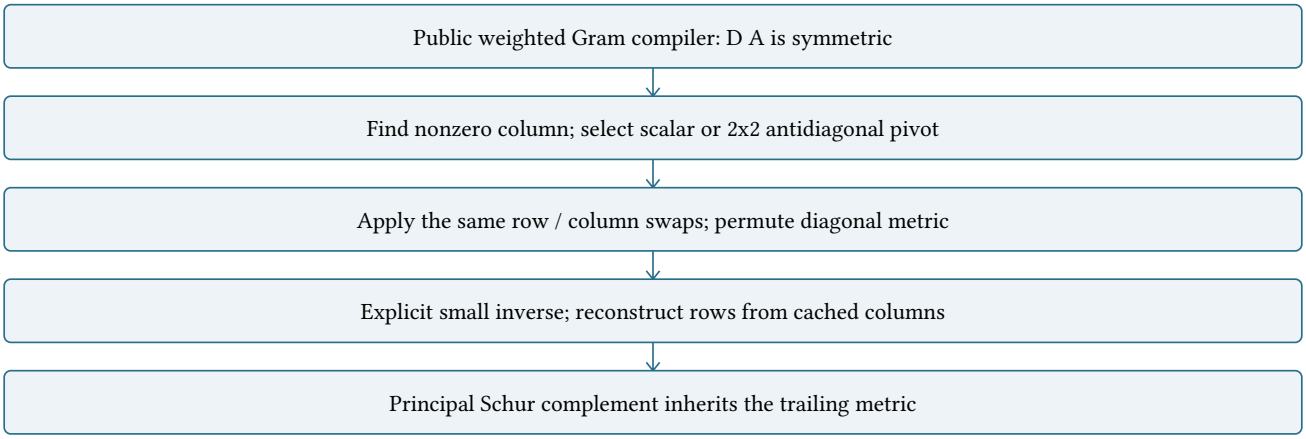
Weighted-symmetry pivot primitives

2026-11-02 · Derived algebra / syntax checks · Numerical trials: zero

Actual compiler. Its weighted Gram matrix B is symmetric and $A=D^{-1}JBJ$, $D=\mathrm{diag}(1,\gamma,\ldots)$. Thus DA is symmetric. The source constructor already returns the inverse diagonal. Finite-field weights need not be positive.



Complete local pivot rule. Find the first nonzero column j. Use its diagonal if nonzero. Otherwise choose i with $A_{ij}\neq 0$ and query column i. The metric gives $d_iA_{ij}=d_jA_{ji}$, so the reverse entry is nonzero. Use diagonal i if nonzero; otherwise the principal pair is (0,u;v,0), with determinant $-uv\neq 0$ and inverse (0,1/v;1/u,0). This costs at most one extra column query. All-zero columns certify a zero source, with RHS compatibility still to be decided.

Closure. Simultaneous permutations give $A'=PAP^{\mathsf{T}}$ and $D'=PDP^{\mathsf{T}}$. Partition $A'=(E,F;C,H)$, $D'=\mathrm{diag}(D_1,D_2)$. For $S=H-CE^{-1}F$, $D_2S=D_2H-F^{\mathsf{T}}D_1E^{-1}F$ is symmetric. The child metric is the trailing portion of D'. It need not remain a geometric progression. Arbitrary separate row and column swaps do not supply this invariant.

Border reuse. A cached column c_j gives row entry $A'_{ji}=d_j^{-1}\,d_i\,c_j(i)$. The helper reconstructs leading rows with zero source queries. It explicitly allocates rows; no complete memory reduction is claimed.

Primitive	Derived accounting
Build metric / inverse	1 inversion + 2(n−1) products
Selection metric validation	n products
Selection queries	j+1 or j+2; n on zero source
Reciprocal-entry guard	2 products on offdiagonal branch
Each simultaneous swap	4 source queries; width +4; 4 products; 2(n+1) additions
Small inverse setup / apply	cut inversions / cut products
Reconstruct cut rows	n+2n·cut products including validation; no query

Evidence scope. Syntax and frozen dependency hashes checked; no numerical, corruption-control or performance trial. Metric inverse checks and the local reciprocal guard do not certify global symmetry. The invariant must come from the public compiler or independent source certification. Cached selection has internal one-use provenance. Existing frozen callers are unchanged.

Next implementation. Complete the symmetry-preserving solve/witness driver, shared weighted block borders and compact block lifts. Preserve zero/singular outcomes and separate feature-exclusion checks. Charge transport queries, changing widths, metric storage and indices. Validate both scalar branches, 2×2 pivots, all permutations, inherited Schur metrics and all upstream cohorts when cost admission permits. No numerical run was launched.

Sources. metric.py, selection.py, transport.py, check_preparation.py and DERIVATION.md. Dumas–Pernet (2018), arXiv:1802.10453, introduction/Section 2, establishes exact-field 1×1/2×2 factorization context. Bostan et al. (2017), DOI 10.1137/16M1062855, provides companion/Schur context. No rank-profile or fast recursive complexity claim is inherited.